

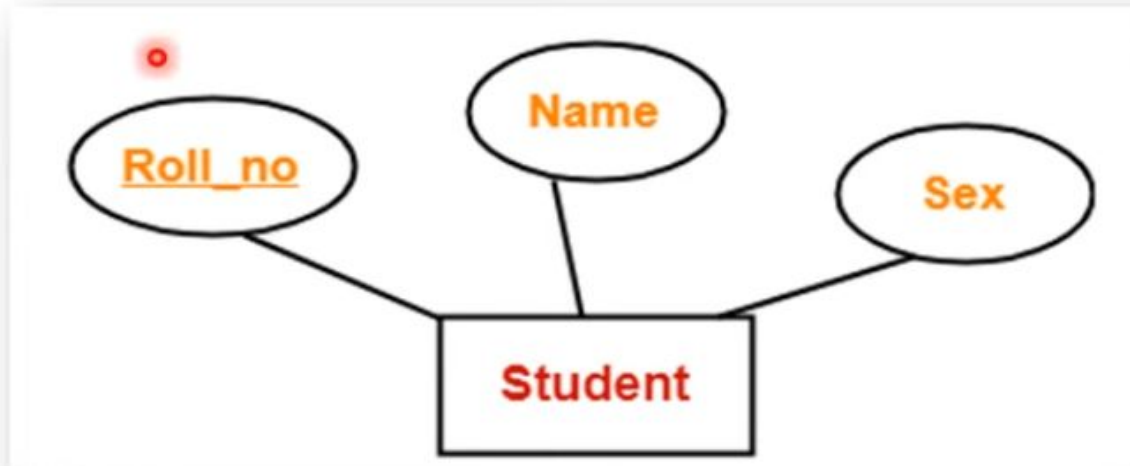
Reduction ER diagram to tables

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Rules of Converting ER Diagram into Table

➤ Rule 1: Strong Entity set with Simple attributes

- Attributes of the table will be the attributes of the entity set.
- The primary key of the table will be the key attribute of the entity set.



<u>Roll_no</u>	Name	Sex

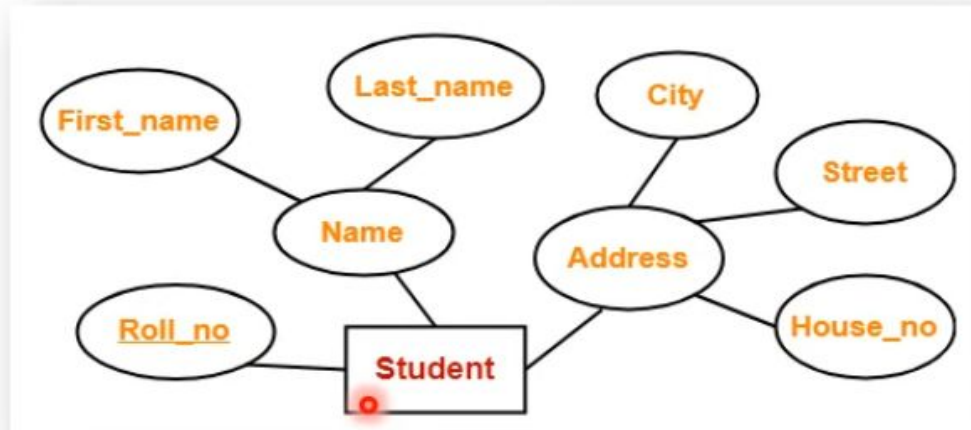
Schema : Student (Roll_no , Name , Sex)

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Rules of Converting ER Diagram into Table

➤ Rule 2: For Strong Entity Set With Composite Attributes

- A strong entity set with any number of composite attributes will require only one table in relational model.
- While conversion, simple attributes of the composite attributes are taken into account and not the composite attribute itself.



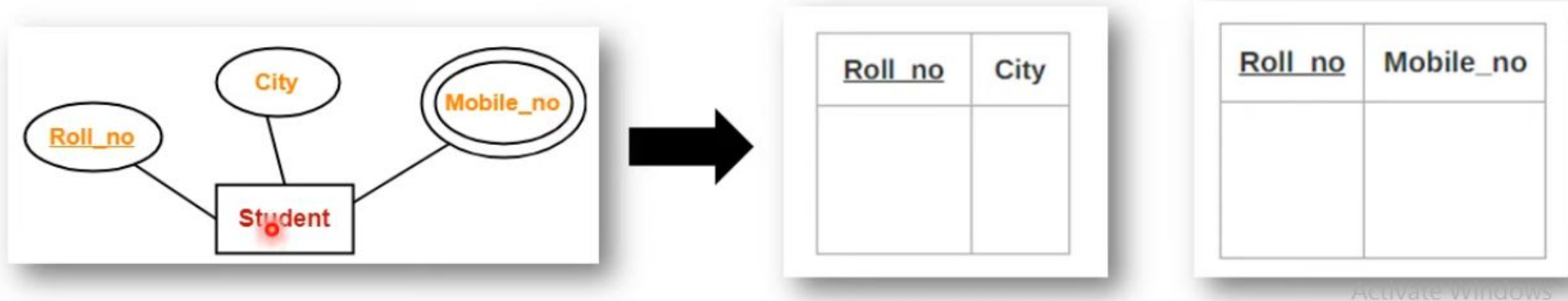
<u>Roll_no</u>	First_name	Last_name	House_no	Street	City

Schema : Student (Roll_no , First_name , Last_name , House_no , Street , City)

Rules of Converting ER Diagram into Table

➤ Rule 3: For Strong Entity Set With Multi Valued Attributes

- A strong entity set with any number of multi valued attributes will require two tables in relational model.
- One table will contain all the simple attributes with the primary key.
- Other table will contain the primary key and all the multi valued attributes.

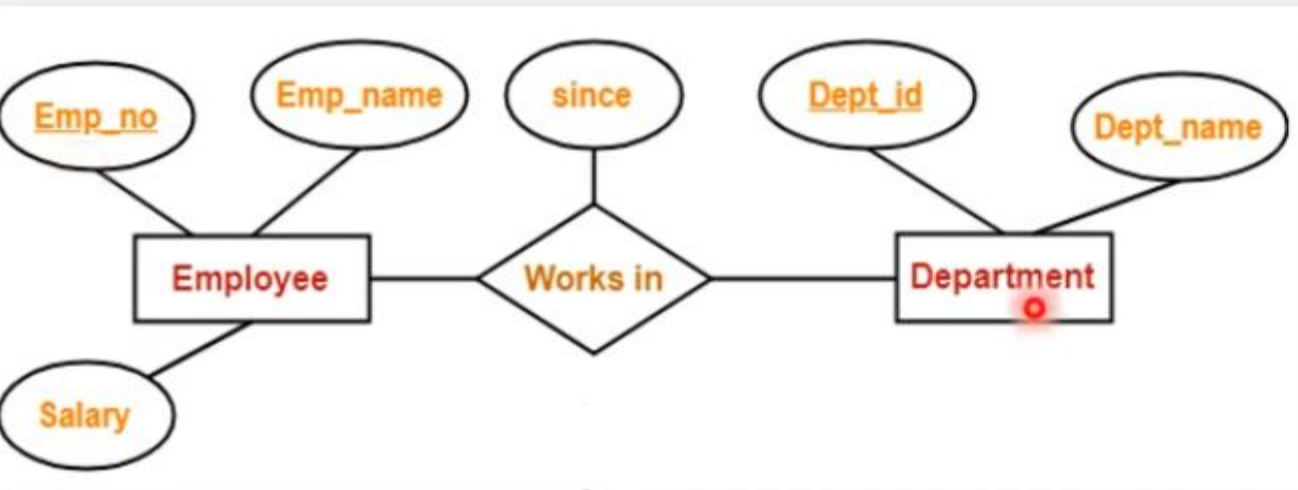


Rules of Converting ER Diagram into Table

➤ Rule 4: Translating Relationship Set into a Table

Attributes of the table are-

- Primary key attributes of the participating entity sets
- Its own descriptive attributes if any.
- Set of non-descriptive attributes will be the primary key.



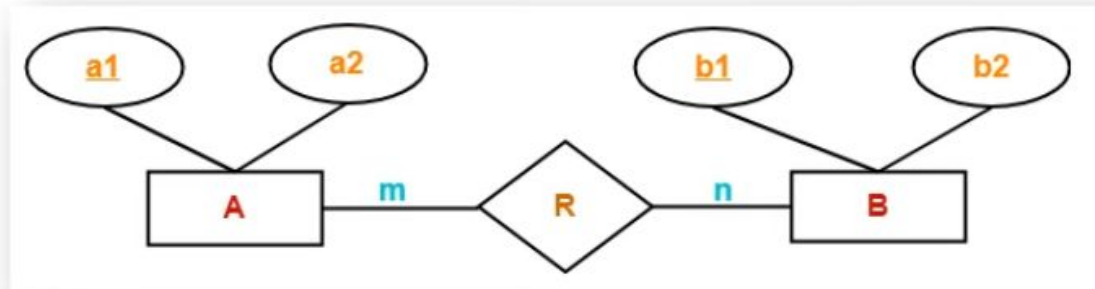
<u>Emp_no</u>	<u>Dept_id</u>	since

Schema : Works in (Emp_no , Dept_id , since)

Rules of Converting ER Diagram into Table

➤ Rule 5: For Binary Relationships With Cardinality Ratios

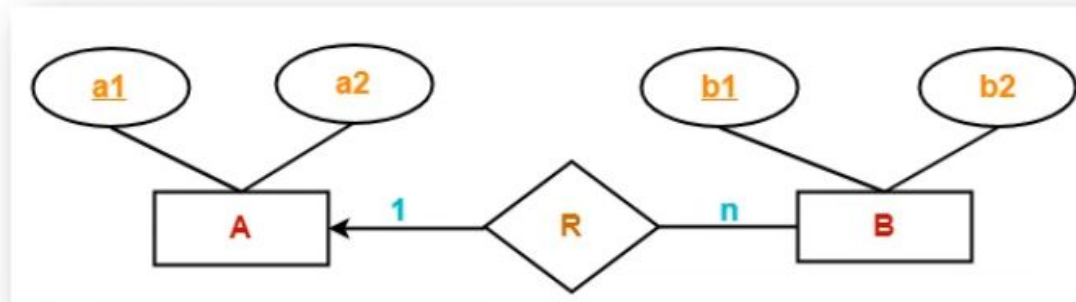
✓ Case-1: For Binary Relationship With Cardinality Ratio m:n



Here, three tables will be required-

1. A (a1 , a2)
2. R (a1 , b1)
3. B (b1 , b2)

✓ Case-2: For Binary Relationship With Cardinality Ratio 1:n



Here, two tables will be required-

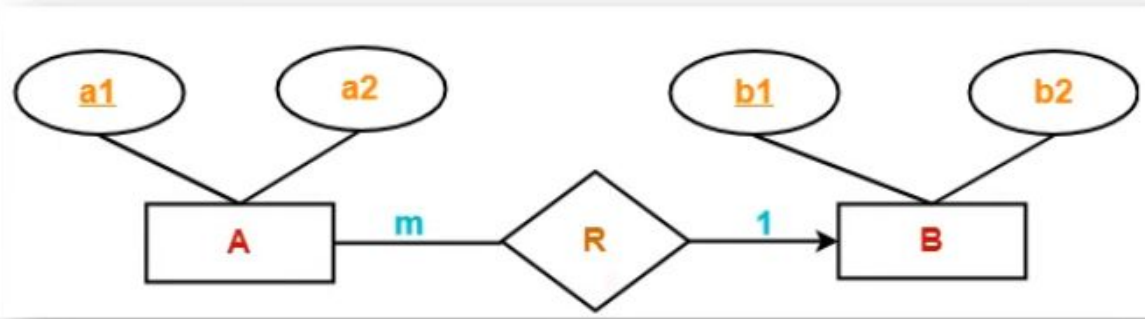
1. A (a1 , a2)
2. BR (a1 , b1 , b2)

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Rules of Converting ER Diagram into Table

➤ Rule 5: For Binary Relationships With Cardinality Ratios

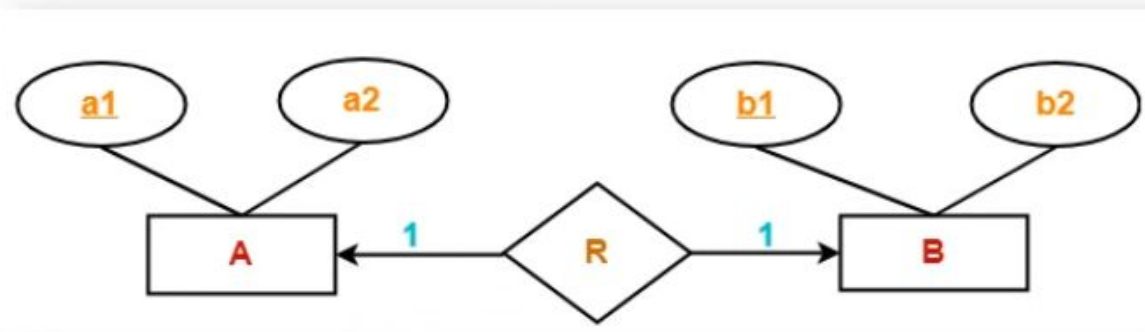
✓ Case-3: For Binary Relationship With Cardinality Ratio m:1



Here, two tables will be required-

1. AR (a1 , a2 , b1)
2. B (b1 , b2)

✓ Case-4: For Binary Relationship With Cardinality Ratio 1:1



Way-01:

1. AR (a1 , a2 , b1)
2. B (b1 , b2)

Way-02:

1. A (a1 , a2)
2. BR (a1 , b1 , b2)

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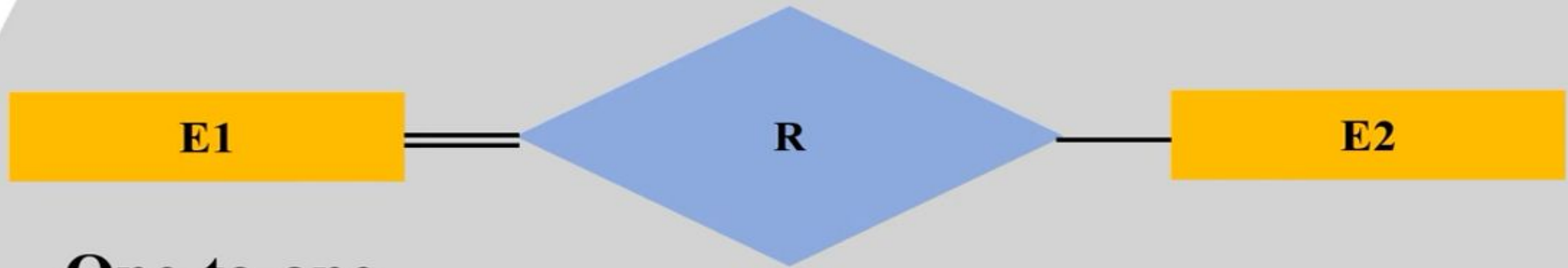
Min, Max Notation

- This is a pair of numbers(m, n) that appear on the connecting line between the entities and their relationships. The minimum number of times an entity can appear in a relation is represented by m whereas, the maximum time it is available is denoted by n .



Each student can enroll minimum **1** and maximum **3** courses, and each course can have maximum **25** students.

Min, Max Notation In Different Types Of Relationship



One-to-one



One-to-many



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Min, Max Notation In Different Types Of Relationship

Many-to-one



Many-to-many

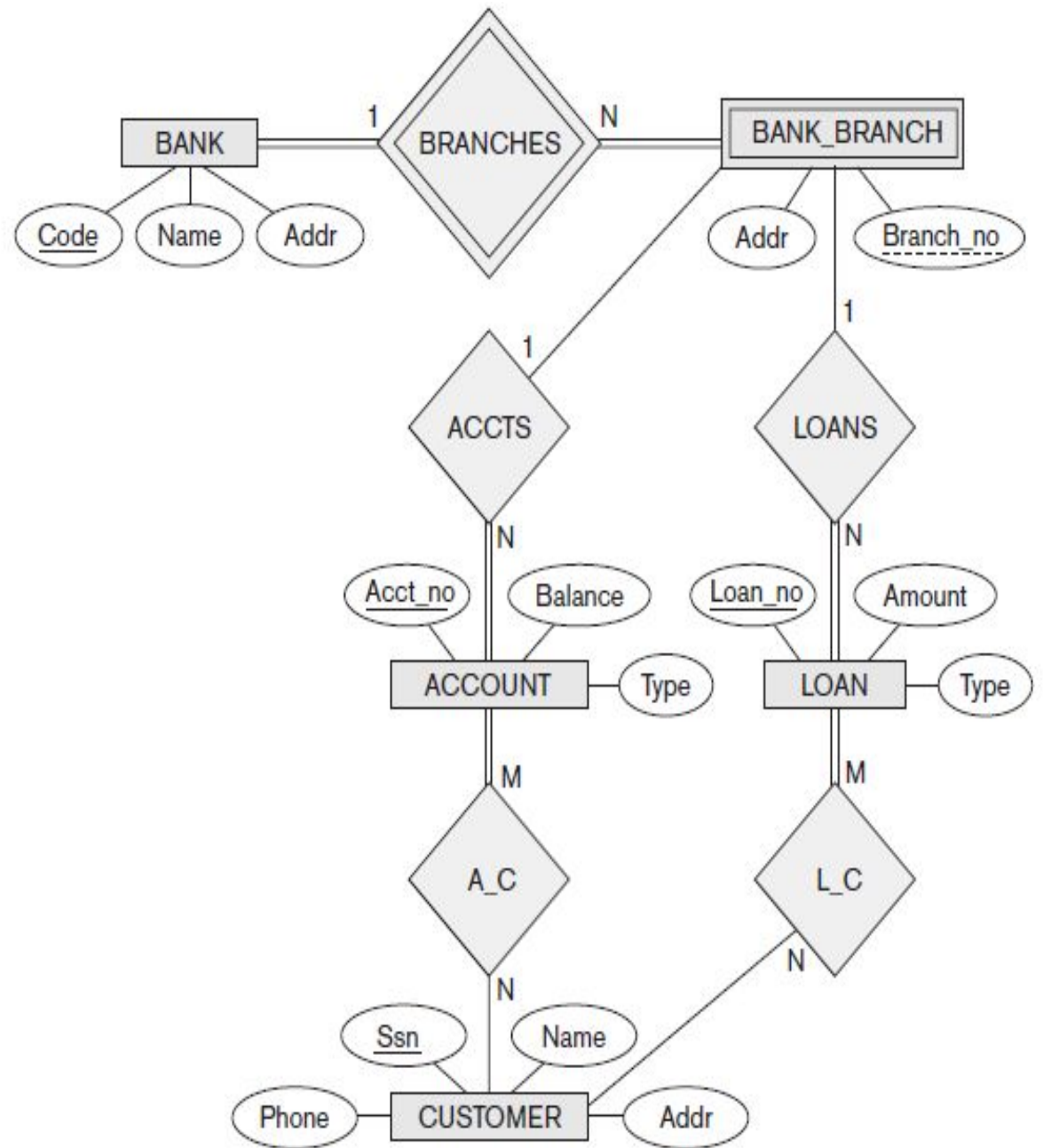


EXERCISES-1

Consider the ER diagram shown in Figure for part of a BANK database.

Each bank can have multiple branches, and each branch can have multiple accounts and loans.

- List the strong (nonweak) entity types in the ER diagram.
- Is there a weak entity type? If so, give its name, partial key, and identifying relationship.
- What constraints do the partial key and the identifying relationship of the weak entity type specify in this diagram?
- List the names of all relationship types, and specify the (min, max) constraint on each participation of an entity type in a relationship type. Justify your choices.

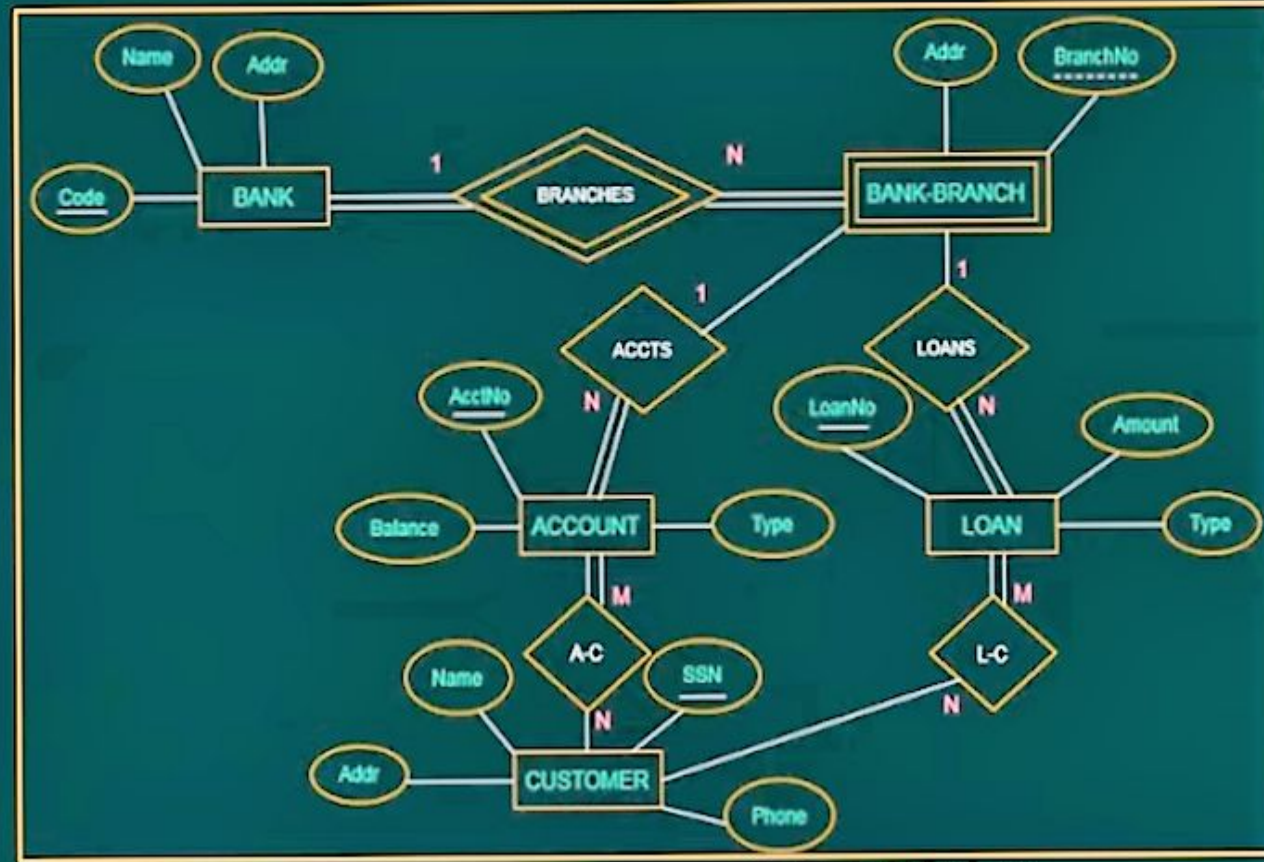


❖ Exercises

1. Consider the ER diagram shown in the figure for part of a BANK database. Each bank can have multiple branches, and each branch can have multiple accounts and loans.

a. List the (nonweak) entity types in the ER diagram.

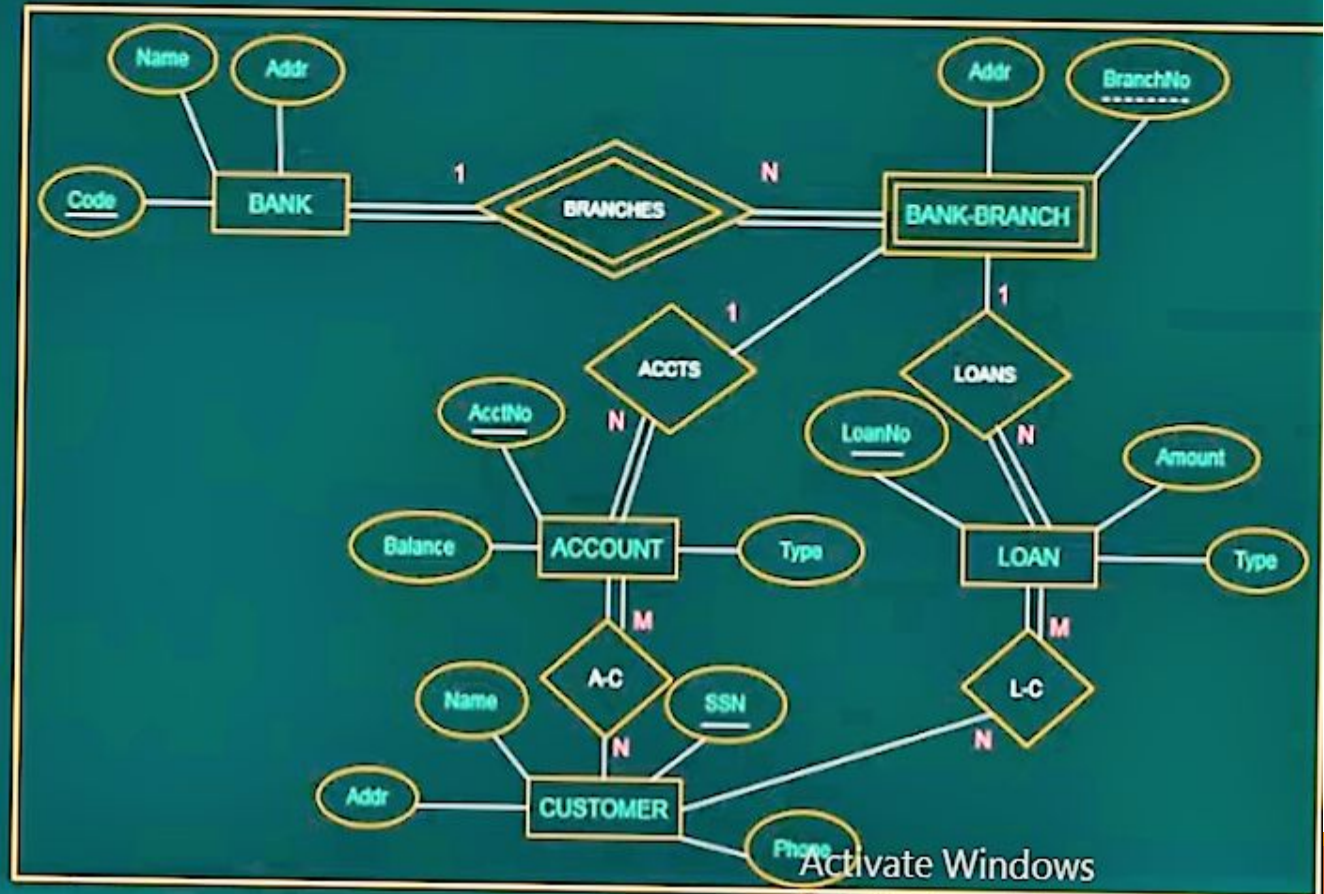
Sol: Bank, Account, Loan, Customer.



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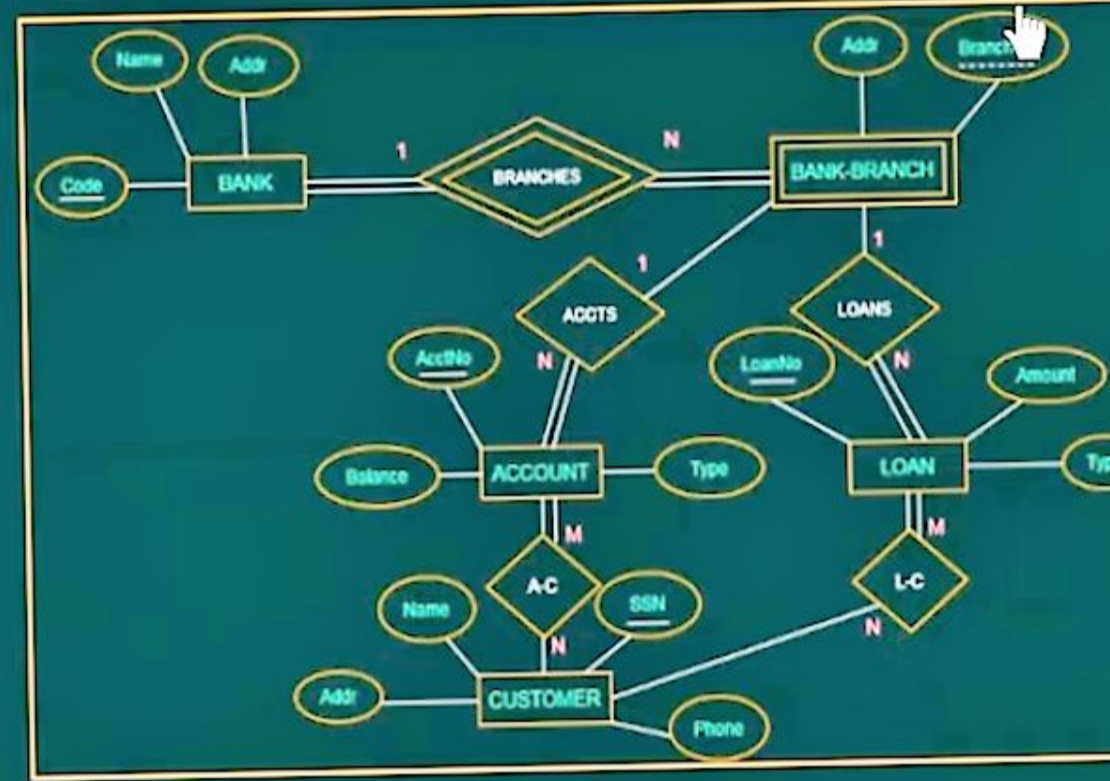
❖ Exercises

b. Is there a weak entity type? If so, give its name, partial key, and identifying relationship.



❖ Exercises

c. What constraints do the partial key and the identifying relationship of the weak entity type specify in this diagram?



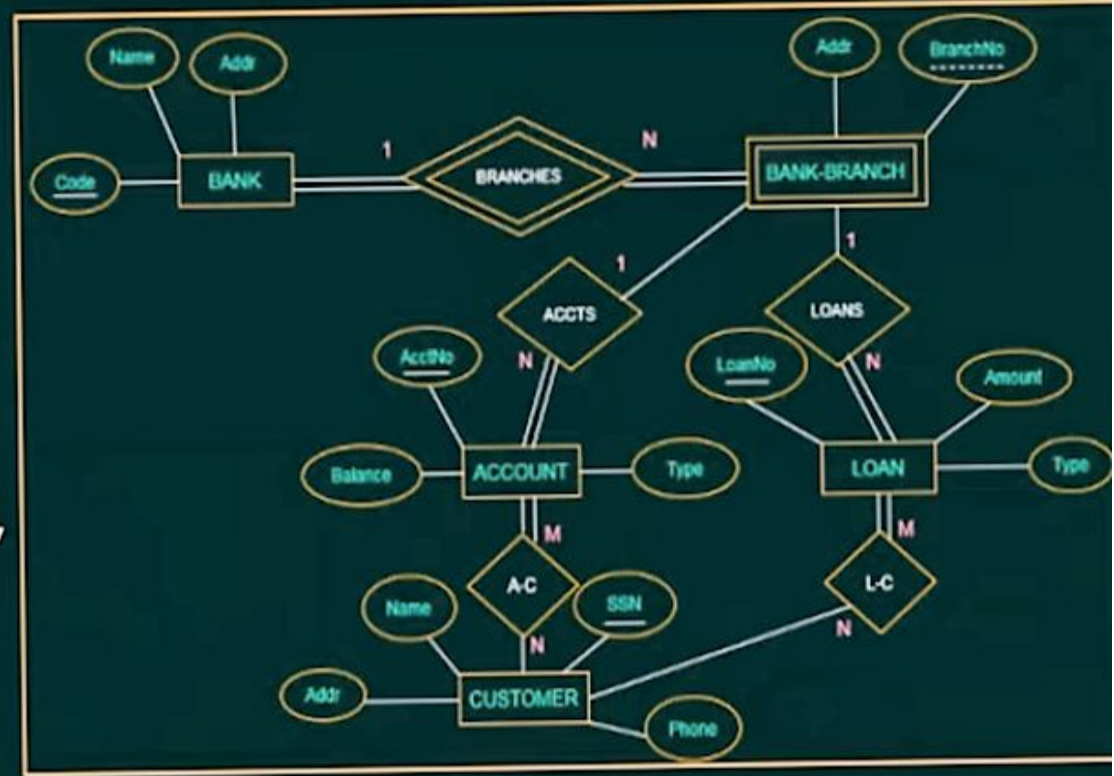
❖ Exercises

c. What constraints do the partial key and the identifying relationship of the weak entity type specify in this diagram?

Sol: The constraint of the partial key BranchNo is that we need to combine BranchNo with Code (owner entity type key), to uniquely identify Bank-Branch.

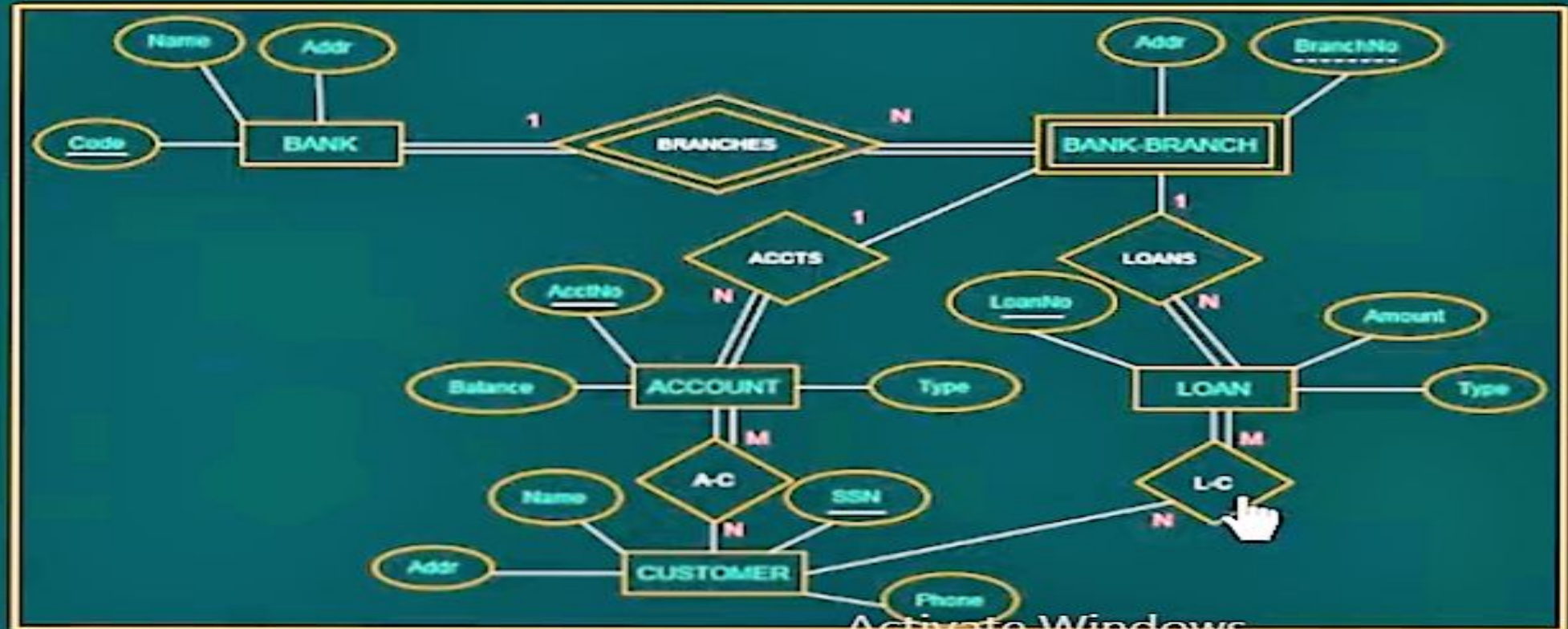
Constraint of the identifying relationship:

(i) The weak entity set must have total participation in the identifying relationship set, Branches.



❖ Exercises

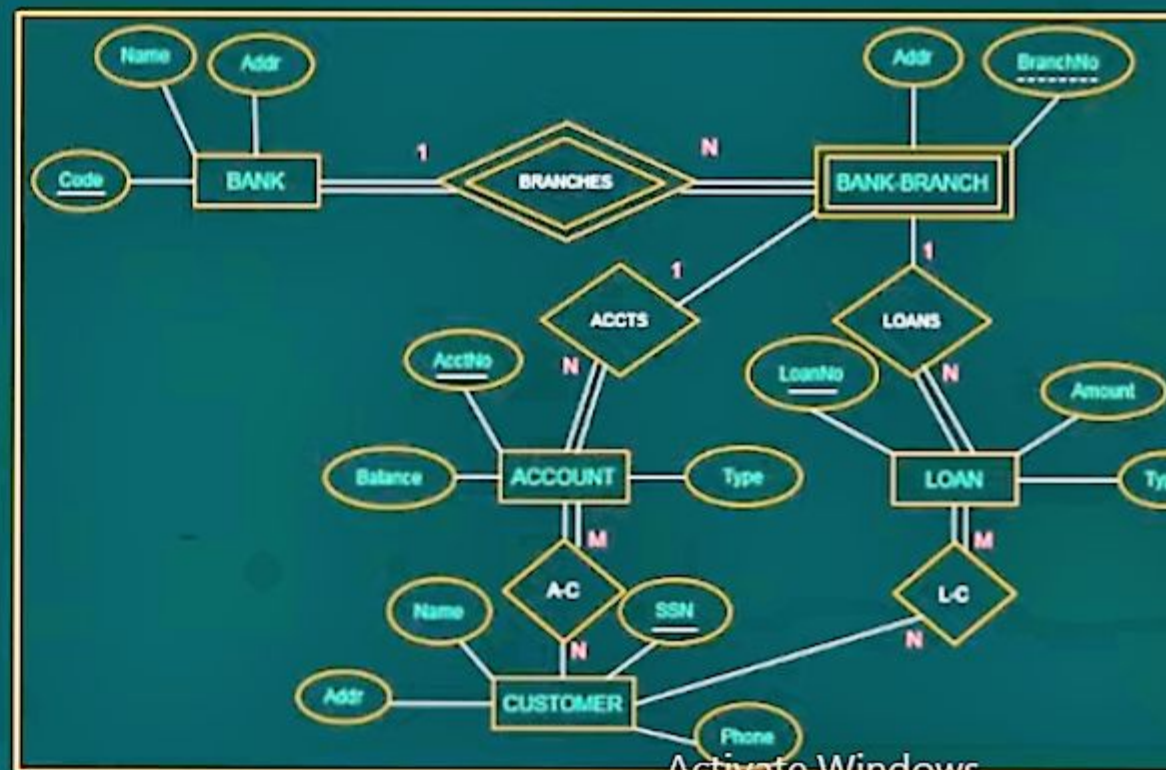
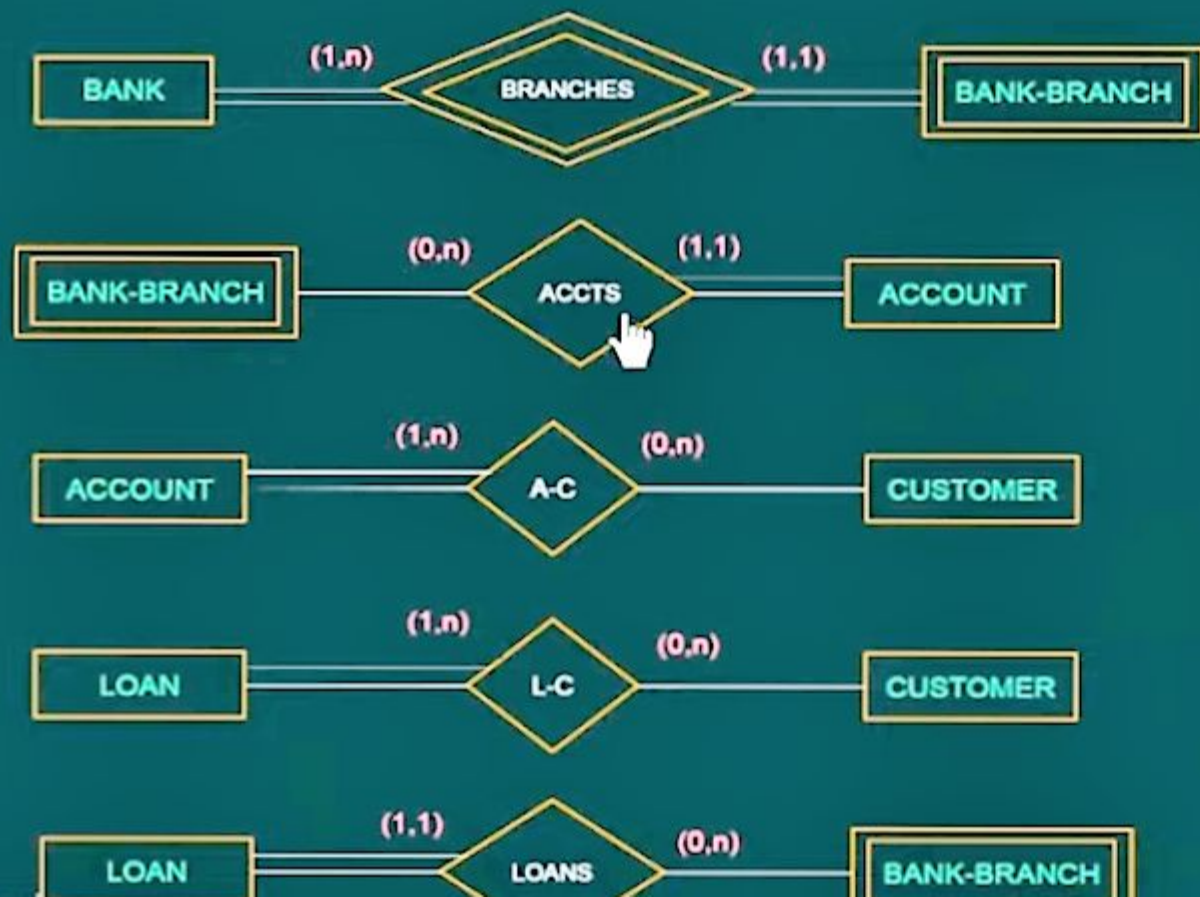
d. List the names of all relationship types, and specify the (min, max) constraint on each participation of an entity type in a relationship type. Justify your choices.



Exercises

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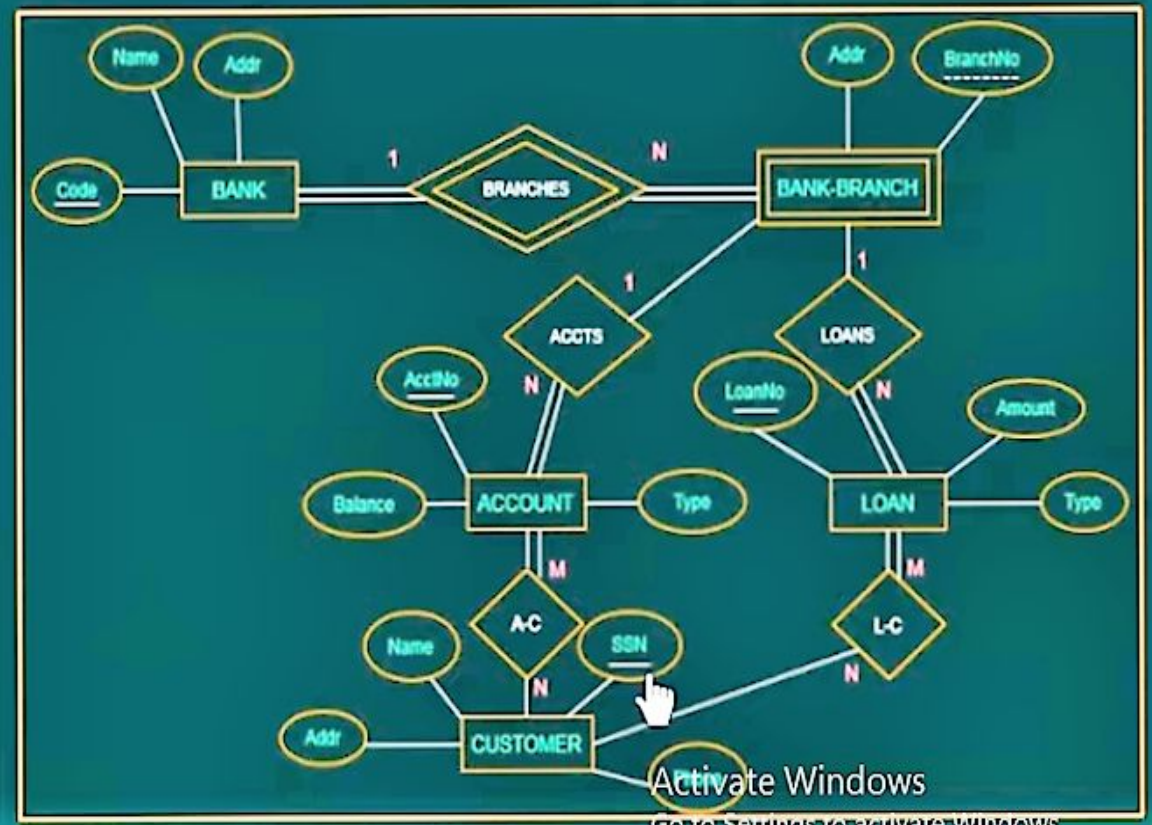
Sol:



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❖ Exercises

e. List concisely the user requirements that led to this ER schema design.

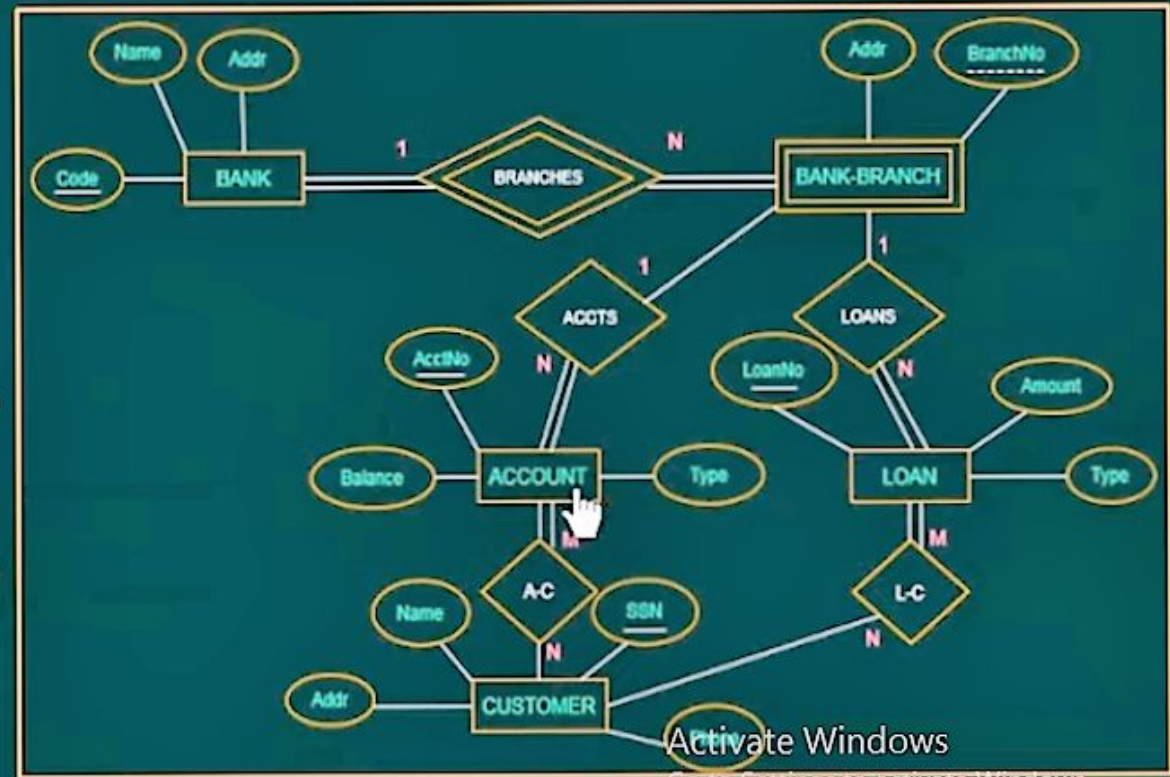


❖ Exercises

e. List concisely the user requirements that led to this ER schema design.

Sol: Requirements can be stated as follows:

- (i) Each Bank has a unique Code, Name and Address.
- (ii) Each Bank has one or more Branches, each of which has a Branch number & address.
- (iii) Each Bank-Branch has zero or more Loans and zero or more accounts.
- (iv) Each account has a unique Account Number, Type & Balance. It is related to exactly one Bank-Branch and to at least one Customer.



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❖ Exercises

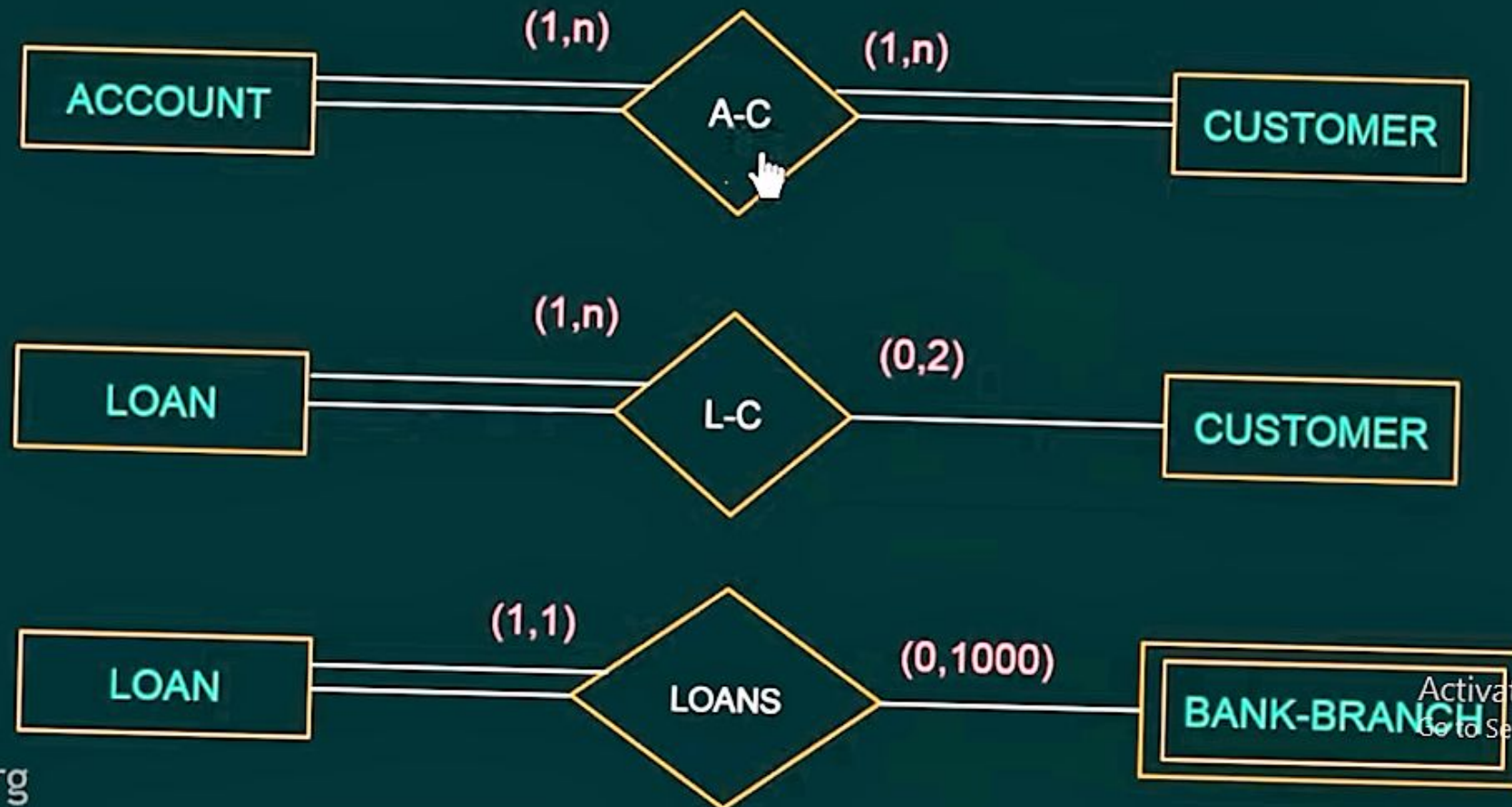
f. Suppose that every customer must have at least one account but is restricted to at most two loans at a time, and that a bank branch cannot have more than 1000 loans. How does this show up on the (min, max) constraints?



❖ Exercises

f. Suppose that every customer must have at least one account but is restricted to at most two loans at a time, and that a bank branch cannot have more than 1000 loans. How does this show up on the (min, max) constraints?

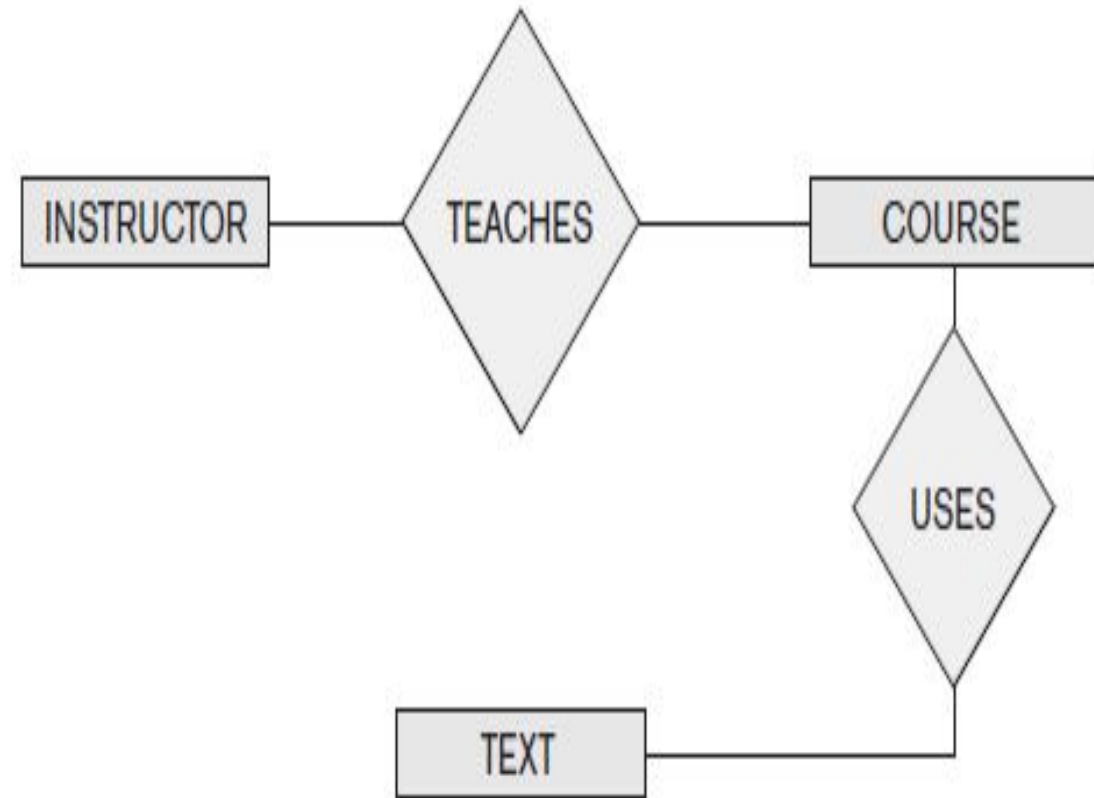
Sol:



EXERCISES-2

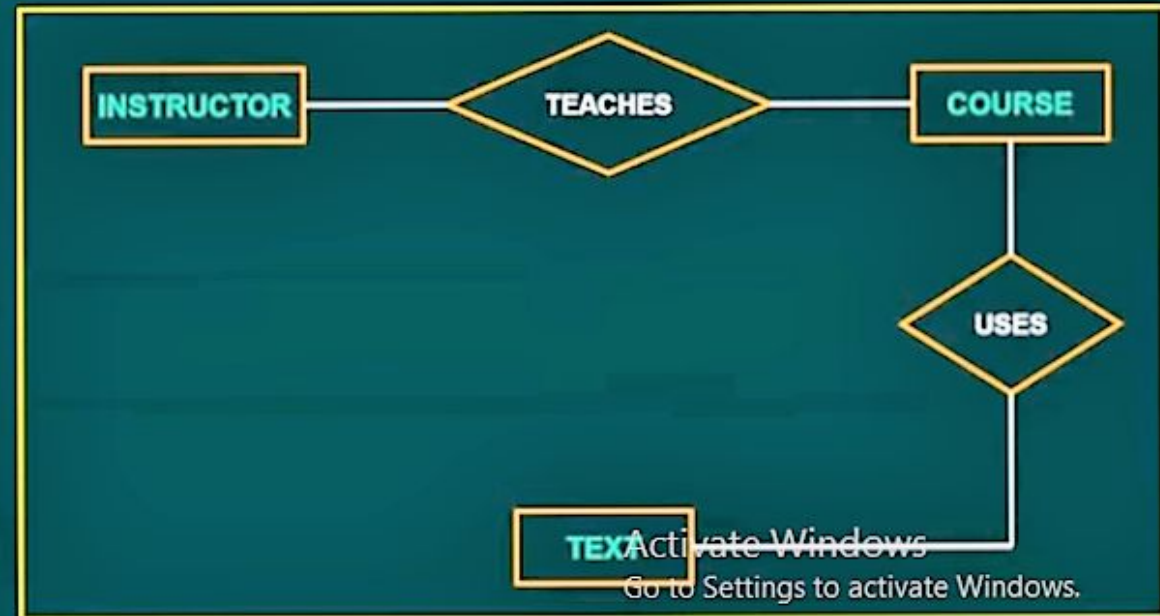
Consider the ER diagram ,

- Assume that a course may or may not use a textbook, but that a text by definition is a book that is used in some course.
- A course may not use more than five books. Instructors teach from two to four courses.
- Supply (min, max) constraints on this diagram.
- *State clearly any additional assumptions you make.*
If we add the relationship ADOPTS, to indicate the textbook(s) that an instructor uses for a course, should it be a binary relationship between INSTRUCTOR and TEXT, or a ternary relationship among all three entity types?



Consider the ER diagram shown in the figure. Assume that a course may or may not use a textbook, but that a text by definition is a book that is used in some course. A course may not use more than five books. Instructors teach from two to four courses.

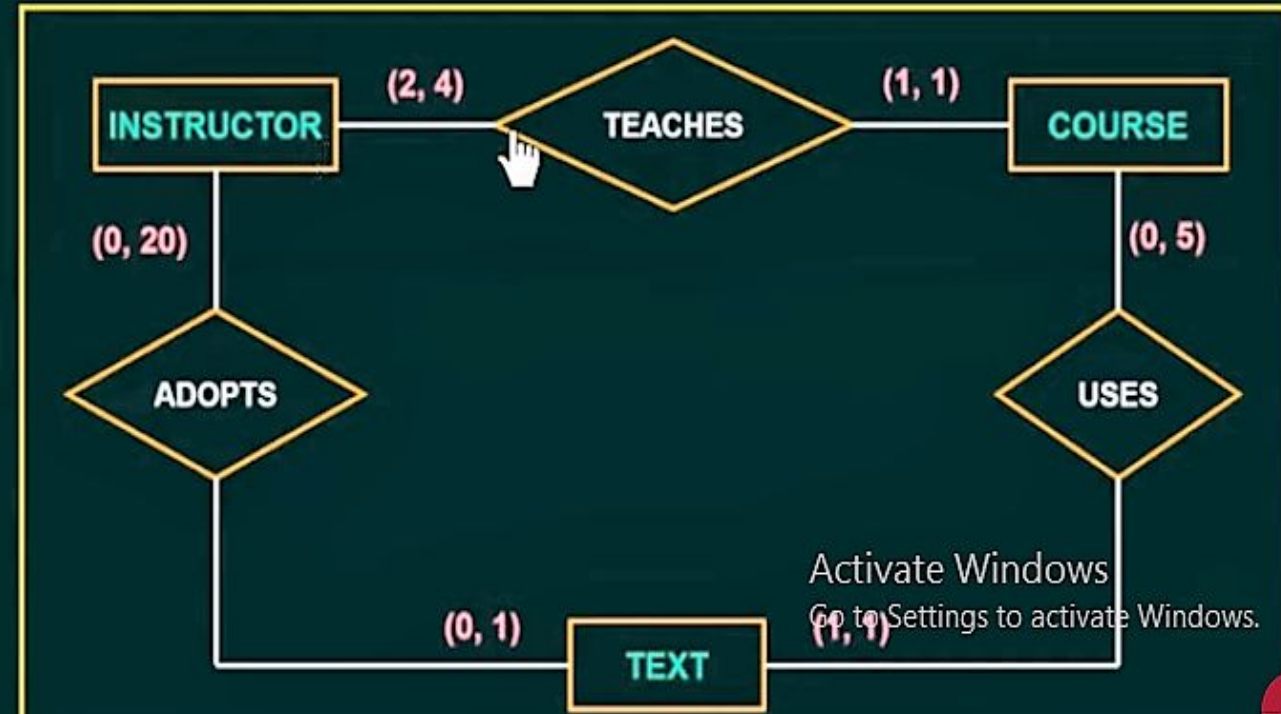
Supply (min, max) constraints on this diagram. State clearly any additional assumptions you make. If we add the relationship ADOPTS between INSTRUCTOR and TEXT, what (min, max) constraints would you put on it? Why?



Sol: Assuming the following additional assumptions:

- (i) Each course is taught by exactly one instructor.
- (ii) Each textbook is used by only one course.
- (iii) An instructor does not have to adopt a textbook for all courses.
- (iv) If a text exists, (a) it is used in some course (b) It is adopted by a instructor who teaches that course.

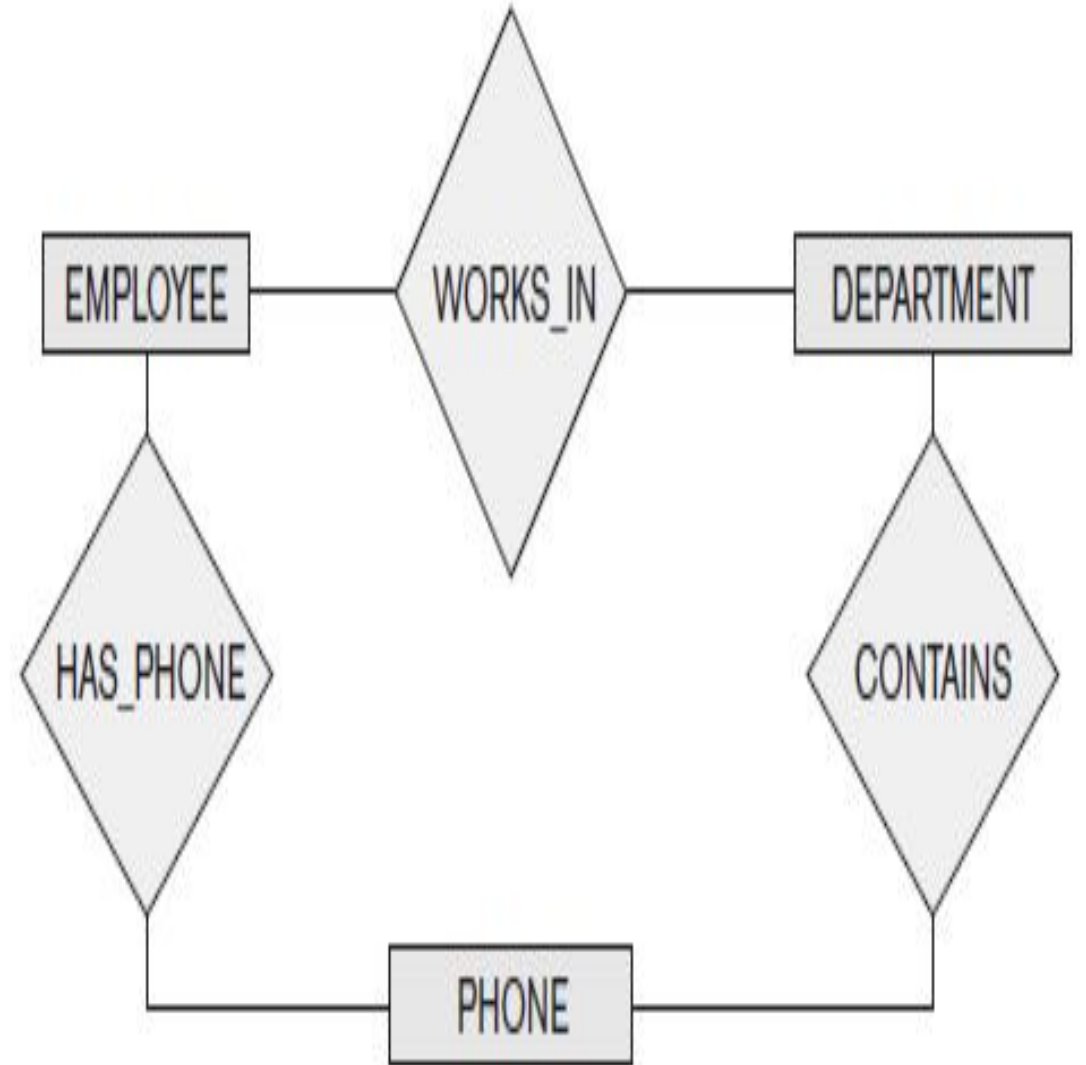
The resulting ER Diagram will have the (min, max) constraints as shown in the figure.



EXCERCISES-3

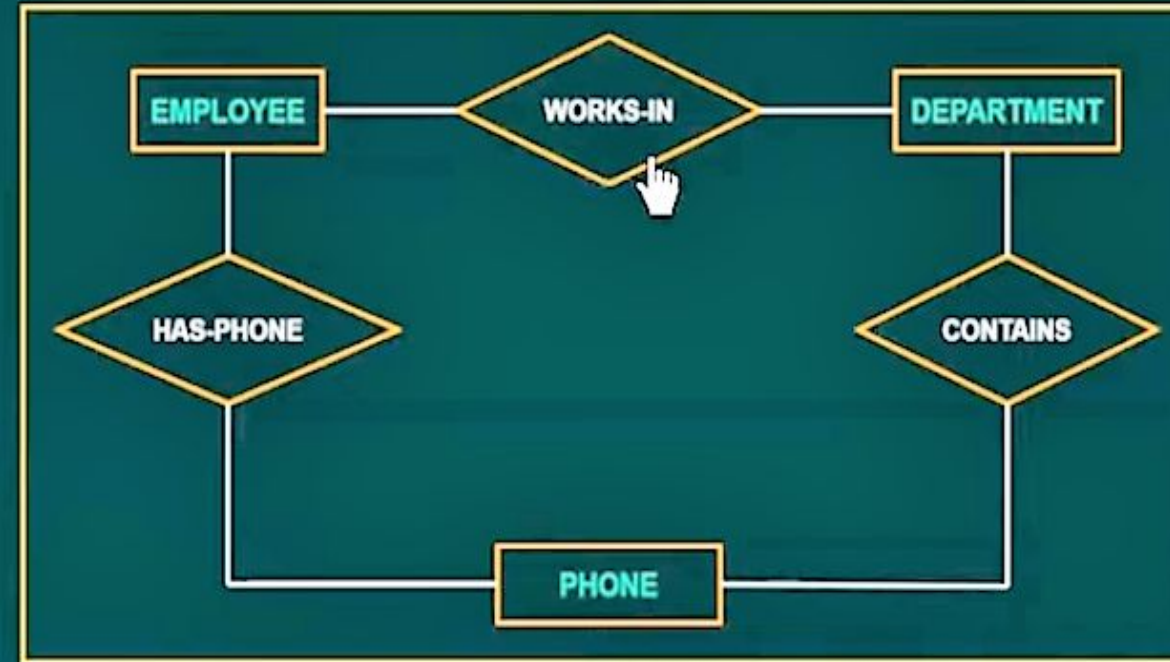
Consider the ER diagram , Assume that an employee may work in up to two departments or may not be assigned to any department.

- Assume that each department must have one and may have up to three phone numbers.
- Supply (min, max) constraints on this diagram.
- *State clearly any additional assumptions you make.*
- Under what conditions would the relationship HAS_PHONE be redundant in this example?



1. Consider the ER diagram in the figure. Assume that an employee may work in upto two departments or may not be assigned to any department. Assume that each department must have one and may have up to three phone numbers.

Supply (min, max) constraints on this diagram. State clearly any additional assumptions you make. Under what conditions would the relationship HAS_PHONE be redundant in this example?

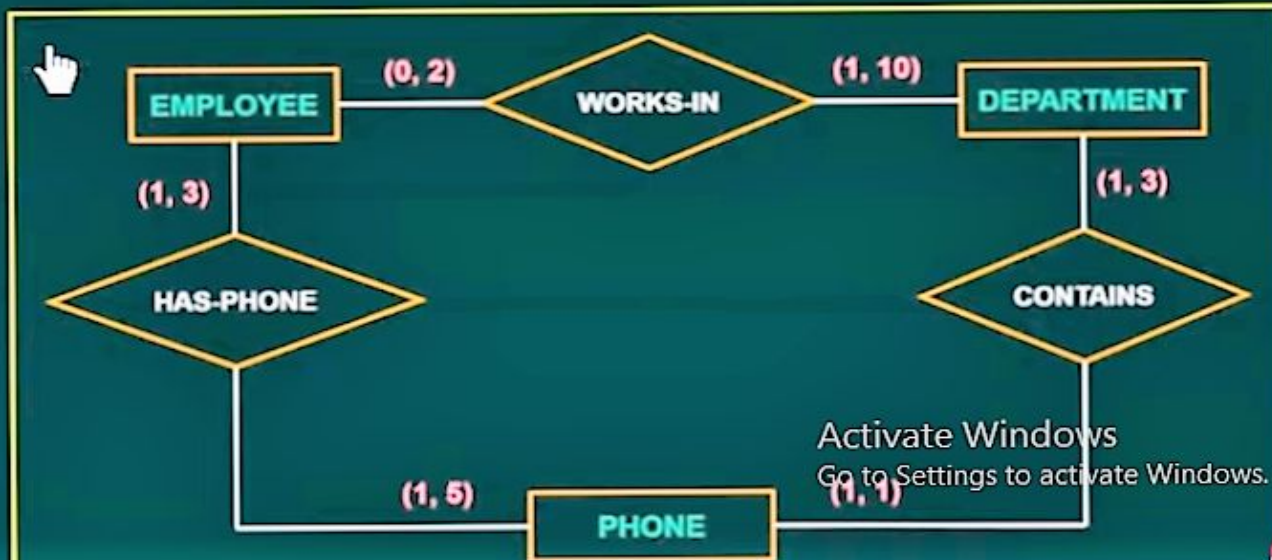


❖ Exercises

Sol: Assuming the following additional assumptions:

- (i) Each department can have between 1 to 10 employees.
- (ii) Each phone is used by only one department.
- (iii) Each phone is assigned to at least one employee and maybe assigned to upto 5 employees.
- (iv) Each employee must have at least one phone and may have upto 3 phones.

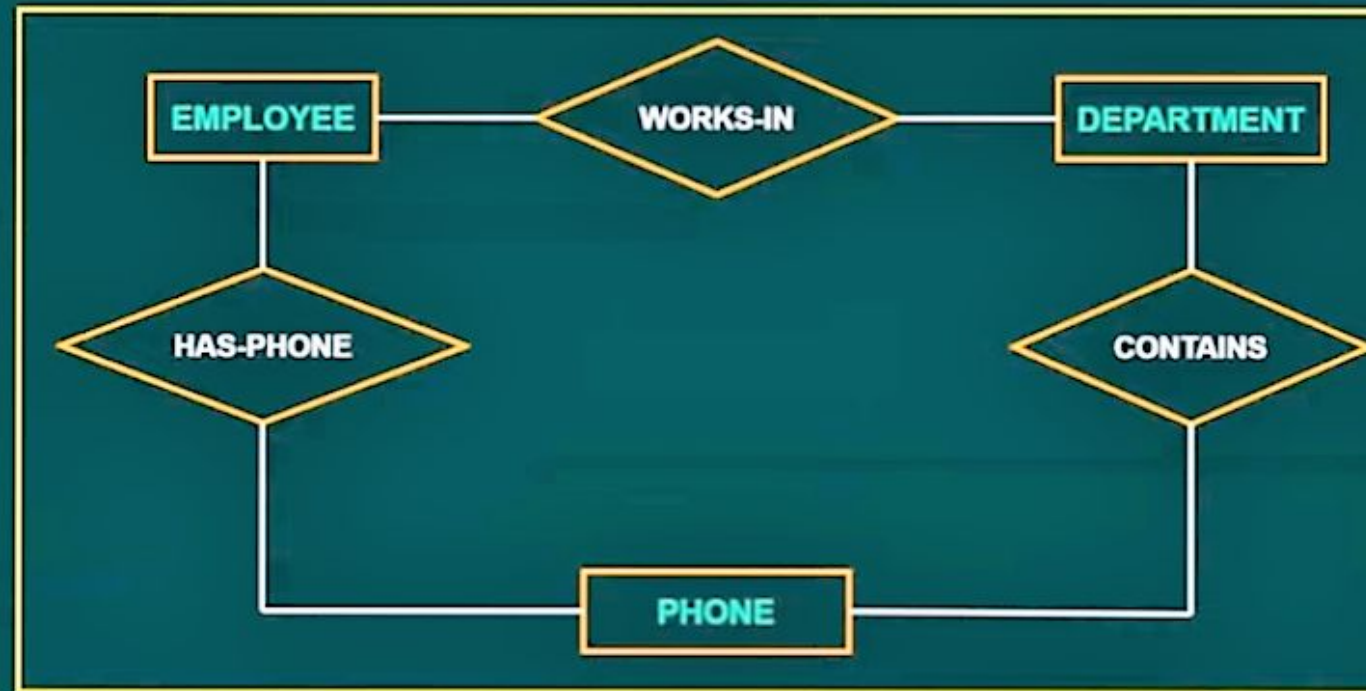
The resulting ER Diagram will have the (min, max) constraints as shown in the figure.



❖ Exercises

Sol: Relationship **HAS-PHONE** would be redundant under the following conditions:

- (i) Each employee is assigned all of the phones of each department that he/she works in.
- (ii) An employee cannot have any other phone outside the department he/she works in.



THANK YOU

